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Complex Numbers

An Original Olympiad Practice Set

compiled & written by Indrajeet Yadav

Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad-style problems where the complex plane does the heavy lifting: roots of unity as a symmetry engine, plane geometry encoded in arithmetic, and trigonometric sums collapsed through the exponential. Each is built to reward a single clean idea rather than computation. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 1/5 · DE MOIVRE, RECIPROCAL SUBSTITUTION, Z + 1/Z = 2COS THETA

A nonzero complex number z satisfies $z + \dfrac{1}{z} = \sqrt{3}$. Without solving for z explicitly, find the value of $z^{12} + \dfrac{1}{z^{12}}$.

2. DIFFICULTY 2/5 · ROOTS OF UNITY, FACTORING Z^N - 1, GEOMETRIC INTERPRETATION OF MODULUS

Twelve points are spaced equally around a circle of radius 2 centered at the origin, forming a regular 12-gon. Pick one vertex V and draw the eleven chords joining V to every other vertex. Find the product of the lengths of these eleven chords.

3. DIFFICULTY 3/5 · ROOTS OF UNITY, PAIRING K WITH N-K, TELESCOPING A SYMMETRIC SUM

Let $\omega = e^{2\pi i/n}$ be a primitive nth root of unity, $n \geq 2$. Evaluate the sum

$$\sum_{k=1}^{n-1} \frac{1}{1-\omega^k},$$

and show the value is real even though the individual terms are not.

4. DIFFICULTY 3/5 · COMPLEX EXPONENTIAL FOR TRIG SUMS, GEOMETRIC SERIES, HALF-ANGLE FACTORING

Show that for every integer $n \geq 1$ and every real θ with $\theta \not\equiv 0 \pmod{2\pi}$,

$$\sum_{k=1}^{n} \sin(k\theta) = \frac{\sin\left(\frac{n\theta}{2}\right) \sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)},$$

by summing a single complex geometric series. Then evaluate $\sum_{k=1}^{17} \sin\left(\frac{k\pi}{9}\right)$, and contrast it with the half-range sum $\sum_{k=1}^{8} \sin\left(\frac{k\pi}{9}\right)$.

5. DIFFICULTY 3/5 · ROTATION BY COMPLEX MULTIPLICATION, EQUILATERAL TRIANGLE CRITERION, PRIMITIVE CUBE ROOT OMEGA

Let a, b, c be the vertices (as complex numbers) of a triangle in the plane. Prove that the triangle is equilateral if and only if

$$a^2 + b^2 + c^2 = ab + bc + ca,$$

and explain geometrically why the condition is exactly $a + \omega b + \omega^2 c = 0$ or $a + \omega^2 b + \omega c = 0$, where $\omega = e^{2\pi i/3}$.

6. DIFFICULTY 4/5 · ROOTS OF UNITY, CSC^2 SUM VIA LOG-DERIVATIVE, LAURENT/TAYLOR EXPANSION AT A ROOT

For an integer $n \geq 2$, prove that

$$\sum_{k=1}^{n-1} \frac{1}{\sin^2\left(\frac{k\pi}{n}\right)} = \frac{n^2-1}{3}.$$

7. DIFFICULTY 4/5 · SUM OF SQUARED DISTANCES, CENTROID IDENTITY, MODULUS EXPANSION

Let $z_1, z_2, \ldots, z_n$ be the nth roots of unity ($n \geq 2$). Compute the sum of the squares of all pairwise distances between them, \[\sum_{1\le j

8. DIFFICULTY 5/5 · CYCLOTOMIC FACTORIZATION, EXPONENT PERMUTATION (COPRIMALITY), PRODUCT OVER ROOTS OF UNITY, BIJECTION OF NONZERO RESIDUES

Let p be a prime with $p \neq 3$, and let $\omega = e^{2\pi i/p}$. Prove that

$$\prod_{k=1}^{p-1} \left(1 + \omega^k + \omega^{2k}\right) = 1.$$

As a concrete instance, deduce that $\displaystyle\prod_{k=1}^{6} \left(1 + \omega^k + \omega^{2k}\right) = 1$ for $\omega = e^{2\pi i/7}$.

Solutions

1. Answer: 2

The hypothesis $z + \frac{1}{z} = \sqrt{3}$ forces $|z| = 1$: writing $z = re^{i\theta}$, the imaginary part of $z + 1/z$ is $(r - r^{-1})\sin\theta$, and since $\sqrt{3}$ is real this must vanish. If $\sin\theta = 0$ then z is real and $z + 1/z = r + 1/r \geq 2 > \sqrt{3}$ (or ≤ -2), impossible; hence $\sin\theta \neq 0$ and $r = r^{-1}$, i.e. $r = 1$. Write $z = e^{i\theta}$. Then $z + \frac{1}{z} = 2\cos\theta = \sqrt{3}$, so $\cos\theta = \frac{\sqrt{3}}{2}$, i.e. $\theta = \pm\frac{\pi}{6}$. The key structural fact is that for $|z| = 1$,

$$z^n + \frac{1}{z^n} = e^{in\theta} + e^{-in\theta} = 2\cos(n\theta).$$

Taking $n = 12$ with $\theta = \pm\frac{\pi}{6}$ gives $n\theta = \pm 2\pi$, hence

$$z^{12} + \frac{1}{z^{12}} = 2\cos(2\pi) = 2.$$

The answer is independent of the sign of θ, as it must be since the two roots are conjugates.

Second method. Recurrence method (no trigonometry). Let $a_n = z^n + z^{-n}$. Then $a_0 = 2$, $a_1 = \sqrt{3}$, and multiplying out $(z^n + z^{-n})(z + z^{-1})$ gives the Chebyshev-type recurrence $a_{n+1} = a_1 a_n - a_{n-1} = \sqrt{3}\, a_n - a_{n-1}$. Computing: $a_2 = 3 - 2 = 1$, $a_3 = \sqrt{3}\cdot 1 - \sqrt{3} = 0$, $a_4 = \sqrt{3}\cdot 0 - 1 = -1$, $a_5 = \sqrt{3}(-1) - 0 = -\sqrt{3}$, $a_6 = \sqrt{3}(-\sqrt{3}) - (-1) = -3 + 1 = -2$. Since $a_6 = z^6 + z^{-6} = -2$ means $z^6 = -1$, we get $z^{12} = 1$ and $a_{12} = 1 + 1 = 2$.

2. Answer: 24576

Place the vertices at $2\omega^k$ for $k = 0, 1, \ldots, 11$, where $\omega = e^{2\pi i/12}$ is a primitive 12th root of unity, and take $V = 2$ (the vertex at $k = 0$). The length of the chord from V to the kth vertex is

$$\left|2 - 2\omega^k\right| = 2\left|1 - \omega^k\right|.$$

The insight is to recognize the numbers ω^k ($k = 0, \ldots, 11$) as exactly the roots of $z^{12} - 1$, so

$$z^{12} - 1 = \prod_{k=0}^{11}(z - \omega^k) = (z - 1)\prod_{k=1}^{11}(z - \omega^k).$$

Dividing by $z - 1$ and letting $z \to 1$,

$$\prod_{k=1}^{11}(1 - \omega^k) = \lim_{z\to 1}\frac{z^{12} - 1}{z - 1} = \left(1 + z + \cdots + z^{11}\right)\Big|_{z=1} = 12.$$

Taking absolute values, $\prod_{k=1}^{11}|1 - \omega^k| = 12$. Therefore the requested product of chord lengths is

$$\prod_{k=1}^{11} 2\,|1 - \omega^k| = 2^{11}\cdot 12 = 2048\cdot 12 = 24576.$$

In general, for a regular n-gon of radius R the product of all chords from one vertex equals $R^{n-1}\,n$.

Second method. Differentiate $P(z) = z^{12} - 1$: since $P(z) = \prod_{k=0}^{11}(z - \omega^k)$, at the simple root $z = 1$ the product rule gives $P'(1) = \prod_{k=1}^{11}(1 - \omega^k)$. But $P'(z) = 12z^{11}$, so $P'(1) = 12$, the same product 12; then multiply by 2^{11} to get 24576.

3. Answer: $\dfrac{n-1}{2}$

Pair the term for k with the term for $n-k$. Since $\omega^{n-k}=\overline{\omega^k}$ (as $|\omega|=1$), the two terms are complex conjugates, so each pair is real; this already proves the sum is real (when n is even the middle term $k=n/2$ gives $\omega^{n/2}=-1$ and $\frac{1}{1-(-1)}=\frac{1}{2}$, itself real). Now compute each pair. Using $\omega^{n-k}=\omega^{-k}$,

$$\frac{1}{1-\omega^k}+\frac{1}{1-\omega^{-k}}.$$

Multiply the second fraction top and bottom by $-\omega^k$:

$$\frac{1}{1-\omega^{-k}}=\frac{-\omega^k}{-\omega^k(1-\omega^{-k})}=\frac{-\omega^k}{-\omega^k+1}=\frac{-\omega^k}{1-\omega^k}.$$

Hence the pair sums to

$$\frac{1}{1-\omega^k}+\frac{-\omega^k}{1-\omega^k}=\frac{1-\omega^k}{1-\omega^k}=1.$$

So every conjugate pair contributes exactly 1. The indices $k=1,\dots,n-1$ split into $\frac{n-1}{2}$ such pairs when n is odd, giving $\frac{n-1}{2}$; when n is even there are $\frac{n-2}{2}$ pairs contributing 1 each plus the lone middle term $\frac{1}{2}$, totaling $\frac{n-2}{2}+\frac{1}{2}=\frac{n-1}{2}$. In both cases

$$\sum_{k=1}^{n-1}\frac{1}{1-\omega^k}=\frac{n-1}{2}.$$

Second method. Logarithmic-derivative method. From $z^n-1=\prod_{k=0}^{n-1}(z-\omega^k)$, divide out the $k=0$ factor: $\frac{z^n-1}{z-1}=1+z+\cdots+z^{n-1}=\prod_{k=1}^{n-1}(z-\omega^k)=:Q(z)$. Then $\sum_{k=1}^{n-1}\frac{1}{z-\omega^k}=\frac{Q'(z)}{Q(z)}$. At $z=1$: $Q(1)=n$ and $Q'(1)=\sum_{j=1}^{n-1}j=\frac{n(n-1)}{2}$, so the sum equals $\frac{n(n-1)/2}{n}=\frac{n-1}{2}$.

4. Answer: Proof of the identity; the full sum $\sum_{k=1}^{17}\sin\left(\frac{k\pi}{9}\right)=0$, while the half-range sum $\sum_{k=1}^{8}\sin\left(\frac{k\pi}{9}\right)=\cot\left(\frac{\pi}{18}\right)$.

Let $z=e^{i\theta}$. The trick is that $\sin(k\theta)=\Im(z^k)$, so the whole sum is the imaginary part of one geometric series:

$$\sum_{k=1}^{n}\sin(k\theta)=\Im\sum_{k=1}^{n}z^k=\Im\left(z\,\frac{z^n-1}{z-1}\right).$$

Now symmetrize by pulling out half-angles. Factor $z^n-1=e^{in\theta/2}\left(e^{in\theta/2}-e^{-in\theta/2}\right)=e^{in\theta/2}\cdot 2i\sin\frac{n\theta}{2}$, and likewise $z-1=e^{i\theta/2}\cdot 2i\sin\frac{\theta}{2}$. Therefore

$$z\,\frac{z^n-1}{z-1}=e^{i\theta}\cdot\frac{e^{in\theta/2}\,2i\sin\frac{n\theta}{2}}{e^{i\theta/2}\,2i\sin\frac{\theta}{2}}=e^{i(n+1)\theta/2}\,\frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}.$$

Taking the imaginary part, $\Im\,e^{i(n+1)\theta/2}=\sin\frac{(n+1)\theta}{2}$, which gives exactly

$$\sum_{k=1}^{n}\sin(k\theta)=\frac{\sin\frac{n\theta}{2}\,\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}},$$

as claimed. (The real part gives the companion cosine sum for free.) Now the numerics, with the deliberate twist that the full sum collapses. Take $\theta=\frac{\pi}{9}$, $n=17$. Then $\frac{(n+1)\theta}{2}=\frac{18}{2}\cdot\frac{\pi}{9}=\pi$, and $\sin\pi=0$, so the formula gives $\sum_{k=1}^{17}\sin\frac{k\pi}{9}=0$. This is no accident: the angles $\frac{k\pi}{9}$ are symmetric about $\frac{9\pi}{9}=\pi$, since term k and term $18-k$ satisfy $\sin\frac{(18-k)\pi}{9}=\sin\left(2\pi-\frac{k\pi}{9}\right)=-\sin\frac{k\pi}{9}$, so they cancel in pairs and the middle $k=9$ gives $\sin\pi=0$. To extract a nonzero value, sum only the first half. With $n=8$, $\theta=\frac{\pi}{9}$ the identity gives

$$\sum_{k=1}^{8}\sin\frac{k\pi}{9}=\frac{\sin\frac{4\pi}{9}\,\sin\frac{\pi}{2}}{\sin\frac{\pi}{18}}=\frac{\sin\frac{4\pi}{9}}{\sin\frac{\pi}{18}}=\frac{\cos\frac{\pi}{18}}{\sin\frac{\pi}{18}}=\cot\frac{\pi}{18},$$

using $\sin\frac{4\pi}{9}=\cos\left(\frac{\pi}{2}-\frac{4\pi}{9}\right)=\cos\frac{\pi}{18}$. So the full $\sum_{k=1}^{17}=0$ while the half-range $\sum_{k=1}^{8}=\cot\frac{\pi}{18}\approx 5.671$.

Second method. Geometric proof of the full-sum cancellation: the points $e^{ik\pi/9}$, $k=0,1,\dots,18$, are the 18th roots of unity (since $e^{i\pi/9}=e^{2\pi i/18}$), and the 18th roots sum to 0. Removing $k=0$ (value 1) and $k=18$ (value 1) leaves $\sum_{k=1}^{17}e^{ik\pi/9}=-2$, which is real, so its imaginary part $\sum_{k=1}^{17}\sin\frac{k\pi}{9}$ is 0, confirming the analytic result.

5. Answer: Proof

Let $\omega = e^{2\pi i/3}$, a primitive cube root of unity, so $1+\omega+\omega^2=0$, $\omega+\omega^2=-1$, and $\omega^3=1$. **The algebraic identity.** Expand the product of the two linear forms, using $\omega^3=1$ wherever a power exceeds 2:

$$(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c) = a^2+b^2+c^2+(\omega+\omega^2)\,(ab+bc+ca).$$

Indeed the squared terms give $a^2+\omega^3 b^2+\omega^3 c^2 = a^2+b^2+c^2$, while each cross product (e.g. the two ways of forming ab) contributes the coefficient $\omega+\omega^2$. Since $\omega+\omega^2=-1$,

$$a^2+b^2+c^2-(ab+bc+ca) = (a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c).$$

Therefore the stated condition $a^2+b^2+c^2=ab+bc+ca$ holds **iff** at least one of the two factors vanishes:

$$a+\omega b+\omega^2 c = 0 \quad \text{or} \quad a+\omega^2 b+\omega c = 0. \tag{†}$$

The geometry. Multiplication by ω is rotation by 120° about the origin. We claim a triangle abc is equilateral with a given orientation iff its third vertex is obtained from the other two by a 60° rotation; concretely, for the counterclockwise orientation,

$$c-a = (b-a)\,e^{i\pi/3}.$$

This single equation says $|c-a|=|b-a|$ and the angle at a is 60°; a triangle with two equal sides enclosing a 60° angle is equilateral, and conversely every counterclockwise equilateral triangle satisfies it. Now $\rho := e^{i\pi/3}$ is a primitive sixth root of unity satisfying $\rho^2-\rho+1=0$. Writing $c-a=\rho(b-a)$ and substituting into $\rho^2-\rho+1=0$ (equivalently, eliminating ρ) yields the polynomial relation

$$(c-a)^2-(c-a)(b-a)+(b-a)^2=0,$$

and expanding this gives exactly $a^2+b^2+c^2-ab-bc-ca=0$. The other orientation $c-a=(b-a)e^{-i\pi/3}$ gives the same final relation. So the two orientations of the equilateral triangle correspond to the two factors in (†). **Equivalence.** If the triangle is equilateral it has one of the two orientations above, hence one factor vanishes, hence $a^2+b^2+c^2=ab+bc+ca$. Conversely, if that identity holds then by (†) one of the oriented 60°-rotation conditions holds, forcing two equal sides with a 60° included angle, i.e. an equilateral triangle. This completes the equivalence.

Second method. Direct side-length verification of the polynomial relation. The minimal relation satisfied by $\rho=e^{i\pi/3}$ is $\rho^2-\rho+1=0$. With $u=b-a$ and $v=c-a$, the equilateral condition $v=\rho u$ gives $v^2-uv+u^2=\rho^2u^2-\rho u^2+u^2=(\rho^2-\rho+1)u^2=0$. Expanding $(c-a)^2-(c-a)(b-a)+(b-a)^2=0$ in a,b,c collapses precisely to $a^2+b^2+c^2-ab-bc-ca=0$, recovering the criterion without the factorization.

6. Answer: Proof

Set $\omega = e^{2\pi i/n}$, so $e^{ik\pi/n}$ has square ω^k. Using $\sin\frac{k\pi}{n} = \frac{e^{ik\pi/n}-e^{-ik\pi/n}}{2i}$,

$$\frac{1}{\sin^2\frac{k\pi}{n}} = \frac{-4}{(e^{ik\pi/n}-e^{-ik\pi/n})^2} = \frac{-4\,\omega^k}{(\omega^k-1)^2}.$$

So we must evaluate $S = -4\sum_{k=1}^{n-1}\frac{\omega^k}{(\omega^k-1)^2}$, where the ω^k ($k=1,\dots,n-1$) are the non-1 nth roots of unity. Split $\frac{r}{(r-1)^2} = \frac{(r-1)+1}{(r-1)^2} = \frac{1}{r-1}+\frac{1}{(r-1)^2}$ with $r=\omega^k$:

$$S = -4\left[\underbrace{\sum_{k=1}^{n-1}\frac{1}{(\omega^k-1)^2}}_{A} + \underbrace{\sum_{k=1}^{n-1}\frac{1}{\omega^k-1}}_{B}\right].$$

From COM-3, $\sum_{k=1}^{n-1}\frac{1}{1-\omega^k} = \frac{n-1}{2}$, so $B = -\frac{n-1}{2}$. For A, use the logarithmic derivative of $Q(z) = 1+z+\cdots+z^{n-1} = \prod_{k=1}^{n-1}(z-\omega^k)$. Then $\frac{Q'}{Q}(z) = \sum_k\frac{1}{z-\omega^k}$ and $\left(\frac{Q'}{Q}\right)'(z) = -\sum_k\frac{1}{(z-\omega^k)^2}$; at $z=1$ this gives $\left(\frac{Q'}{Q}\right)'(1) = -A$. To find $\left(\frac{Q'}{Q}\right)'(1)$, expand Q about $z=1$ via $z=1+t$: since $Q(z) = \frac{z^n-1}{z-1}$ and $(1+t)^n-1 = nt+\binom{n}{2}t^2+\binom{n}{3}t^3+\cdots$,

$$Q(1+t) = n+\binom{n}{2}t+\binom{n}{3}t^2+\cdots = n\Big(1+\tfrac{n-1}{2}t+\tfrac{(n-1)(n-2)}{6}t^2+\cdots\Big).$$

Then $\ln Q = \ln n+\ln\big(1+u\big)$ with $u = \frac{n-1}{2}t+\frac{(n-1)(n-2)}{6}t^2+\cdots$; using $\ln(1+u) = u-\frac{u^2}{2}+\cdots$,

$$\ln Q = \ln n+\tfrac{n-1}{2}t+\Big(\tfrac{(n-1)(n-2)}{6}-\tfrac{1}{2}\big(\tfrac{n-1}{2}\big)^2\Big)t^2+\cdots.$$

The coefficient of t is $\frac{Q'}{Q}(1) = \frac{n-1}{2}$ (consistent with COM-3), and the coefficient of t^2 equals $\frac{1}{2}\big(\frac{Q'}{Q}\big)'(1)$. Hence

$$-A = \left(\frac{Q'}{Q}\right)'(1) = 2\left(\frac{(n-1)(n-2)}{6}-\frac{(n-1)^2}{8}\right) = \frac{(n-1)(n-2)}{3}-\frac{(n-1)^2}{4},$$

so

$$A = \frac{(n-1)^2}{4}-\frac{(n-1)(n-2)}{3} = \frac{(n-1)\big[3(n-1)-4(n-2)\big]}{12} = \frac{(n-1)(5-n)}{12}.$$

Finally

$$S = -4(A+B) = -4\left(\frac{(n-1)(5-n)}{12}-\frac{n-1}{2}\right) = -\frac{(n-1)(5-n)-6(n-1)}{3} = -\frac{(n-1)(-1-n)}{3} = \frac{(n-1)(n+1)}{3} = \frac{n^2-1}{3}.$$

Thus $\sum_{k=1}^{n-1}\csc^2\frac{k\pi}{n} = \dfrac{n^2-1}{3}$, as required.

Second method. Chebyshev sanity / induction check. The numbers $\cos\frac{k\pi}{n}$ ($k=1,\dots,n-1$) are the roots of the Chebyshev polynomial U_{n-1} of the second kind, and $\csc^2\frac{k\pi}{n} = \frac{1}{1-\cos^2\frac{k\pi}{n}}$ is a symmetric function of those roots; expressing the sum through the coefficients of U_{n-1} reproduces $\frac{n^2-1}{3}$. A quick check: $n=2$ gives $\csc^2\frac{\pi}{2} = 1 = \frac{4-1}{3}$; $n=3$ gives $2\csc^2\frac{\pi}{3} = 2\cdot\frac{4}{3} = \frac{8}{3} = \frac{9-1}{3}$; $n=4$ gives $2\csc^2\frac{\pi}{4}+\csc^2\frac{\pi}{2} = 4+1 = 5 = \frac{16-1}{3}$.

7. Answer: n^2 for the roots of unity; in general $n\sum_j |z_j|^2$ when the centroid is at the origin.

Work with ordered pairs first. Expand the modulus squared:

$$|z_j - z_k|^2 = (z_j - z_k)\overline{(z_j - z_k)} = |z_j|^2 + |z_k|^2 - z_j\bar{z}_k - \bar{z}_j z_k.$$

Sum over **all ordered pairs** (j,k), $1 \le j,k \le n$ (the diagonal $j = k$ contributes 0):

$$\sum_{j,k} |z_j - z_k|^2 = \sum_{j,k}\big(|z_j|^2 + |z_k|^2\big) - \sum_{j,k}\big(z_j\bar{z}_k + \bar{z}_j z_k\big).$$

The first part is $2n\sum_j |z_j|^2$ (each $|z_j|^2$ appears n times as $|z_j|^2$ and n times as $|z_k|^2$). The second part factors as

$$\sum_{j,k} z_j\bar{z}_k = \Big(\sum_j z_j\Big)\Big(\sum_k \bar{z}_k\Big) = \Big|\sum_j z_j\Big|^2,$$

and similarly for the conjugate term, so the subtracted quantity is $2\big|\sum_j z_j\big|^2$. Therefore

$$\sum_{j,k} |z_j - z_k|^2 = 2n\sum_j |z_j|^2 - 2\Big|\sum_j z_j\Big|^2.$$

This is the master identity. If the centroid is at the origin, $\sum_j z_j = 0$, the second term vanishes, and the unordered sum is half of the ordered one: \[\boxed{\sum_{j

Second method. Per-vertex symmetry. By rotational symmetry, $\sum_{k\ne j} |z_j - z_k|^2$ is the same for every fixed j; call it C. For $z_j = 1$, $\sum_{k=1}^{n-1} |1 - \omega^k|^2 = \sum_{k=1}^{n-1}\big(2 - 2\cos\frac{2\pi k}{n}\big) = 2(n-1) - 2\,\Re\sum_{k=1}^{n-1}\omega^k = 2(n-1) - 2(-1) = 2n$. So $C = 2n$, the total over ordered pairs is $nC = 2n^2$, and the unordered sum is n^2.

8. Answer: Proof (value $= 1$)

The engine is the factorization $1 + x + x^2 = \dfrac{x^3 - 1}{x - 1}$, valid whenever $x \ne 1$. For each $k \in \{1, \dots, p-1\}$, $\omega^k \ne 1$, so

$$1 + \omega^k + \omega^{2k} = \frac{\omega^{3k} - 1}{\omega^k - 1}.$$

Taking the product,

$$\prod_{k=1}^{p-1}\big(1 + \omega^k + \omega^{2k}\big) = \frac{\displaystyle\prod_{k=1}^{p-1}\big(\omega^{3k} - 1\big)}{\displaystyle\prod_{k=1}^{p-1}\big(\omega^k - 1\big)}. \tag{$*$}$$

The decisive observation is about the **exponents in the numerator**. As k runs through $\{1, 2, \dots, p-1\}$, the residues $3k \bmod p$ also run through $\{1, 2, \dots, p-1\}$ — a permutation. This holds precisely because $\gcd(3,p) = 1$ (guaranteed by p prime and $p \ne 3$): multiplication by 3 is a bijection of the nonzero residues modulo p. Since ω^{3k} depends only on $3k \bmod p$ (as $\omega^p = 1$),

$$\prod_{k=1}^{p-1}\big(\omega^{3k} - 1\big) = \prod_{j=1}^{p-1}\big(\omega^j - 1\big),$$

the **same** set of factors as the denominator, just reordered. The numerator and denominator of $(*)$ are therefore equal (both nonzero, since $\omega^j - 1 \ne 0$ for $1 \le j \le p-1$), so

$$\prod_{k=1}^{p-1}\big(1 + \omega^k + \omega^{2k}\big) = 1.$$

Why must $p \ne 3$? If $p = 3$ then ω is a primitive cube root of unity and the factor at $k = 1$ is $1 + \omega + \omega^2 = 0$, so the product collapses to 0 and the bijection breaks ($3k \equiv 0$). For every other prime the value is exactly 1. For $p = 7$: $3 \cdot \{1,2,3,4,5,6\} \equiv \{3,6,2,5,1,4\} \pmod 7$, which is $\{1,2,3,4,5,6\}$ rearranged, so numerator and denominator cancel termwise and $\prod_{k=1}^{6}(1 + \omega^k + \omega^{2k}) = 1$.

Second method. Cyclotomic-polynomial viewpoint. Let $\Phi(z) = 1 + z + \cdots + z^{p-1} = \prod_{k=1}^{p-1}(z - \omega^k)$. Each desired factor is $g(\omega^k)$ with $g(x) = 1 + x + x^2 = (x - \rho_1)(x - \rho_2)$, where $\rho_{1,2} = e^{\pm 2\pi i/3}$. Then $\prod_{k=1}^{p-1} g(\omega^k) = \prod_k(\omega^k - \rho_1)(\omega^k - \rho_2) = \Phi(\rho_1)\Phi(\rho_2)$ up to the sign $(-1)^{2(p-1)} = 1$. Now $\Phi(\rho) = \frac{\rho^p - 1}{\rho - 1}$ for $\rho \ne 1$. Since $\rho_{1,2}$ are primitive cube roots and $\gcd(p,3) = 1$, the map $\rho \mapsto \rho^p$ permutes $\{\rho_1, \rho_2\}$; hence $\Phi(\rho_1)\Phi(\rho_2) = \frac{(\rho_1^p - 1)(\rho_2^p - 1)}{(\rho_1 - 1)(\rho_2 - 1)}$ has equal numerator and denominator, giving 1.